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Euclidean Geometry In Mathematical Olympiads

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Problem 1.39 CGMO 2012 P5



CGMO

incircle

Chapter 1

fgm

fgmcon



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AlastorMoody

2163 posts

Nov 29, 2018, 2:45 PM • 4

#1

Problem 1.39 (CGMO 2012/5) Let $\triangle ABC$ be a triangle. The incircle of $\triangle ABC$ is tangent to AB and AC at D and E respectively. Let O denote the circumcenter of $\triangle BCI$. Prove that $\angle ODB = \angle OEC$.

AlastorMoody

2163 posts

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#2

Solution



By Incenter- Excenter Lemma,

 $A - I - O$, By SAS Congruency, $\triangle ADO \cong \triangle AEO \implies \angle ODB = \angle OEC$

kootrapali

4527 posts

Feb 29, 2020, 6:27 PM • 1

#5

Solution



Notice that $\angle ICB = \frac{1}{2}\angle C$ and $\angle IBC = \frac{1}{2}\angle B$ since I is the incenter. Let K be the point diametrically opposite I on the circumcircle of $\triangle BCI$. Then $\angle IKB = \angle ICB = 90 - \angle BIK$ and $\angle IKC = \angle IBC = 90 - \angle CIK$.

Thus, $\angle BIK = 90 - \frac{1}{2}\angle C$ and $\angle CIK = 90 - \frac{1}{2}\angle B$. Also,

$\angle EIC = 90 - \frac{1}{2}\angle C$ and $\angle DIB = 90 - \frac{1}{2}\angle B$, so

$\angle EIO = \angle DIO = 180 - \frac{1}{2}(\angle B + \angle C)$. By SAS congruence, we have $\triangle EIO$ congruent to $\triangle DIO$, or $\angle IEO = \angle IDO$. Since $\angle IEC = \angle IDB = 90^\circ$, we have $\angle ODB = \angle OEC$.

This post has been edited 1 time. Last edited by kootrapali, Jun 8, 2020, 2:21 PM

Archimedes15

1490 posts

Jun 8, 2020, 1:49 PM • 2

#7



AlastorMoody wrote:

Solution

I wish I noticed this solution. Literally a one-liner.

However, this is what I figured out when I tried to solve the problem:

Solution



As $\angle IEC = \angle IDB = 90$, it suffices to show that $\angle IEO = \angle ODI$.

IDE is isosceles so it suffices to show that DEO is isosceles.

DEO has to be isosceles because DAE is isosceles and IDE is isosceles implying that the angle bisector of $\angle DAE$ is the perpendicular bisector of segment DE.

Since DEO and IDE are both isosceles, $\angle IEO = \angle IDO$ and because of the first sentence of this solution, $\angle ODB = \angle OEC$

I think this works.

Jerry122805

1074 posts

Aug 12, 2020, 7:45 PM • 1 🍌

#8

Solution



By incenter-excenter O is on (ABC) but that doesn't really matter. Now observe $AD = AE$ by two tangent and $\angle DAO = \angle EAO$ since angle bisector. Then, we know that $\triangle DAO \cong \triangle EAO$ so $\angle AEO = \angle AEO$ and thus $\angle ODB = \angle OEC$.

Zanderman

4375 posts

Mar 16, 2021, 1:59 PM • 2 🍌

#10

I can't believe this problem.

I spent like 3 days completely stuck on it. -- I instantly noticed the Incenter-Excenter lemma, but then missed the triangle congruence. And then today after school, I spent like a second